A Proofs

A.1 Proof of Theorem 1

We first define the decision version of Problem 1. Given an instance $(I, \Sigma, \{P_i\}, \{A_i^j\}, \{F_j\})$ and a threshold η, the problem asks whether there exists a repaired trajectory $I' = (S', X')$ such that $I' \models \Sigma$ and $\sum_{j=1}^{m} \sum_{i=1}^{n} \left(x'_{ij} - F_j(s'_i)\right)^2 \leq \eta$, where $s'_i \in P_i$ and $x'_{ij} \in A_i^j$ for all repaired entries.

We first show that the decision version belongs to NP. Given a candidate repair I', we can verify in polynomial time that (i) each repaired position and attribute value belongs to its candidate set, (ii) all area consistency constraints in Σ are satisfied by evaluating $\Delta A_{ij} \leq \delta_j$ for every attribute j and every consecutive triplet, and (iii) the objective value can be computed and compared with η. Hence, the decision version is in NP.

We next prove NP-hardness by a polynomial-time reduction from 3-SAT. Consider a 3-SAT instance with variables $U = \{u_1, \ldots, u_p\}$ and clauses $C_1, \ldots, C_q$, where each clause contains exactly three literals. We construct a restricted instance of Problem 1 as follows.

We consider a single attribute ($m = 1$) and fix all spatial positions, i.e., $P_i = \{s_i\}$ for every point i, so that only attribute values are subject to repair. This restricts the general joint-repair problem to a special case; thus, NP-hardness of this restricted version implies NP-hardness of the original problem. For each variable occurrence, we introduce a trajectory point with candidate set $\{-1, +1\}$, encoding Boolean values False and True, respectively. We further fix the neighborhood position-attribute model to a constant function $F_1(s_i) = 0$ for all i.

By fixing spatial displacements to a constant, the area consistency constraint reduces to a local form: $|2x_i - x_{i-1} - x_{i+1}| \leq \delta$, which captures dependencies among consecutive points. Under this construction, all feasible assignments incur an objective value bounded by a polynomial in n, and by setting η sufficiently large, the decision problem depends only on the satisfaction of the area consistency constraints.

We then construct a set of local gadgets to encode the logical relations of the formula. For each variable u_k, we build a variable gadget that connects all its occurrences through auxiliary points, such that the induced constraints enforce a consistent assignment $x_i \in \{-1, +1\}$ across all occurrences. For each clause $C_\ell = (y_1 \vee y_2 \vee y_3)$, we construct a clause gadget consisting of a constant number of auxiliary points, such that a constraint of the form $g(y_1, y_2, y_3) \leq \delta$ is satisfied if and only if at least one literal takes value +1. Such constructions can be realized by appropriately fixing spatial displacements and candidate values so that constraint violations precisely correspond to unsatisfied clauses.

Formally, let ϕ be a truth assignment of the 3-SAT instance. We construct a corresponding repair $I'(\phi)$ satisfying Σ. Conversely, any feasible repair I' induces a truth assignment $\phi(I')$ that satisfies all clauses. Therefore, the 3-SAT instance is satisfiable if and only if the constructed instance admits a feasible repair.

Since the reduction is polynomial in the size of the formula, the decision version of Problem 1 is NP-hard. Combined with the fact that it belongs to NP, the decision version is NP-complete. Since the decision version reduces to the optimization version, the optimization version of Problem 1 is NP-hard.

A.2 Proof of Proposition 2

According to Algorithm 1, the complexity of the exact algorithm TJER consists of two parts. First, in Line 1, DPMinRepair enumerates minimal repair states for each attribute by scanning the trajectory sequentially. Since each point admits at most four local repair decisions, namely repairing neither, the position only, the attribute only, or both, the number of repair states for one attribute is bounded by 4^n in the worst case, yielding a complexity of $O(n4^n)$. Repeating this process for all m attributes gives $O(mn4^n)$. Second, in Line 4–Line 8, the algorithm performs an exhaustive search over the candidate value space for each repair structure. In the worst case, all n positions and all mn attribute values are repaired, and each repaired variable has at most K candidates. Therefore, the size of the search space is bounded by $K^n \cdot K^{mn} = K^{(m+1)n}$. For each candidate assignment, checking feasibility and computing the objective require at most $O(mn)$ time, so this stage costs $O(mnK^{(m+1)n})$. Combining the two parts, the total time complexity is $O(mn4^n) + O(mnK^{(m+1)n}) = O\Big(mn(4^n + K^{(m+1)n})\Big)$.

A.3 Proof of Proposition 3

According to Algorithm 2, the complexity of TJCR consists of two parts. First, for each attribute j, Lines 2–4 detect the violating triplets, select one responsible point for each triplet, and construct the corresponding minimal point-level repair sets. Since the number of violating triplets is at most linear in the trajectory length n, and the subsequent coverage checking and repair-set construction over the candidate points take $O(n^2)$ time in the worst case, this stage costs $O(mn^2)$ over all m attributes. Second, in Lines 7–9, the algorithm sequentially selects repair values for each repaired point. For every point i, the position candidate set $\mathcal{P}_i$ and the attribute candidate set $\mathcal{A}_i^j$ both have size K, so the joint search over $(s_i, x_{ij}) \in \mathcal{P}_i \times \mathcal{A}_i^j$ requires $O(K^2)$ time. Since at most n points are repaired for each attribute, this stage costs $O(mnK^2)$ in total. Combining the two parts, TJCR returns the approximate repair solution in $O(mn^2 + mnK^2)$ time.

A.4 Proof of Proposition 4

Consider the j-th attribute at the i-th point. Let the observation be x_{ij} and let $\hat{x}_{ij}$ denote the model-estimated value of attribute j at position s_i. The overall objective is the sum of normalized squared errors:

$$L = \sum_{j=1}^{m} \frac{1}{\sigma_j^2} \sum_{i=1}^{n} \left(\hat{x}_{ij} - x_{ij}\right)^2.$$

Let $L^\star$ denote the optimal loss achieved by the exact algorithm.

Define the per-point normalized loss for the approximate and exact solutions as

$$\ell_i^{(j)} = \frac{\left(\hat{x}'_{ij} - x'_{ij}\right)^2}{\sigma_j^2}, \qquad \ell_i^{(j)\star} = \frac{\left(\hat{x}^\star_{ij} - x^\star_{ij}\right)^2}{\sigma_j^2},$$

where $\hat{x}'_{ij}$ and $\hat{x}^\star_{ij}$ denote the model-estimated values at the repaired positions s'_i and $s^\star_i$, respectively. By the bounded-residual condition, for all i, j,

$$\ell_i^{(j)} \leq \frac{M^2}{\sigma_j^2}, \qquad \ell_i^{(j)\star} \leq \frac{M^2}{\sigma_j^2}.$$

Fix an arbitrary pair (i, j) and compare $\ell_i^{(j)}$ with $\ell_i^{(j)\star}$. If the approximate solution repairs this point while the exact solution does not, then by the residual bound we obtain

$$\ell_i^{(j)} - \ell_i^{(j)\star} \;\le\; \frac{M^2}{\sigma_j^2}.$$

Otherwise, both solutions repair this point but choose different repairs. The exact algorithm jointly searches over the position and attribute candidate sets and selects the best pair $(s_i^\star, x_{ij}^\star)$ from K^2 combinations, while TJCR selects (s_i', x_{ij}') from a restricted candidate space. Since both residuals are bounded by M, in the worst case they can have opposite signs, which maximizes the difference between squared errors. Thus we have

$$\left(\hat{x}_{ij}' - x_{ij}'\right)^2 - \left(\hat{x}_{ij}^\star - x_{ij}^\star\right)^2 \;\le\; (2M)^2 \;=\; 4M^2.$$

Dividing both sides by σ_j^2 gives

$$\ell_i^{(j)} - \ell_i^{(j)\star} \;\le\; \frac{4M^2}{\sigma_j^2}.$$

Finally, this provides a uniform upper bound on the per-point loss gap (and dominates the previous case), hence for all i, j,

$$\ell_i^{(j)} - \ell_i^{(j)\star} \;\le\; \frac{4M^2}{\sigma_j^2}.$$

Summing over $i = 1, \ldots, n$ and $j = 1, \ldots, m$ yields

$$L - L^\star \;\le\; \sum_{j=1}^{m} \frac{4nM^2}{\sigma_j^2}.$$

A.5 Proof of Proposition 5

According to Algorithm 3, we analyze the complexity by examining each step for a fixed attribute j in Line 1. In Line 2, DetectViolations scans the trajectory once and computes all violating triplets, which takes $O(n)$ time. Lines 3–6 perform a single pass over the trajectory to detect points that violate two adjacent constraints; since each point participates in a constant number of triplets, this step also costs $O(n)$. Next, Lines 7–9 iterate over the set of violating triplets V_j, whose size is at most $O(n)$, and for each triplet perform a constant-time check and selection, leading to another $O(n)$ time. Finally, Lines 10–13 process each repair point in R_j, where $|R_j| \le n$. For each point i, the algorithm computes $\bar{s}_i$ and $\bar{x}_{ij}$ in constant time, and evaluates a constant number of repair modes, namely position-only, attribute-only, and joint repair, to minimize the neighborhood position-attribute model loss; thus this step also costs $O(3n)$. Summing all steps, the total cost for one attribute is $O(3n)$, and over all m attributes, the total time complexity is $O(3mn)$.

A.6 Proof of Proposition 6

We use the same objective and per-point loss definitions as in the proof of Proposition 4. In particular,

$$L \;=\; \sum_{j=1}^{m} \frac{1}{\sigma_j^2} \sum_{i=1}^{n} \left(\hat{x}_{ij} - x_{ij}\right)^2,$$

and

$$\ell_i^{(j)} \;=\; \frac{\left(\hat{x}_{ij}' - x_{ij}'\right)^2}{\sigma_j^2}, \qquad \ell_i^{(j)\star} \;=\; \frac{\left(\hat{x}_{ij}^\star - x_{ij}^\star\right)^2}{\sigma_j^2}.$$

Fix (i, j) and compare $\ell_i^{(j)}$ with $\ell_i^{(j)\star}$. If only the approximate solution repairs this point, then

$$\ell_i^{(j)} - \ell_i^{(j)\star} \;\le\; \frac{M^2}{\sigma_j^2}.$$

Otherwise, both solutions repair point i but produce different repairs. Let $(s_i^\star, x_{ij}^\star)$ and (s_i', x_{ij}') denote the exact and TJGR repairs, respectively. Under TJGR, the repair is determined by midpoint anchors from neighboring clean points, while the exact solution lies within a feasible region of radius θ. Since the midpoint anchor averages the deviations of two neighboring clean points and each side may differ from the optimal local repair by at most θ together with an additional half-step offset, this midpoint operation may carry over deviations from both sides rather than canceling them. Therefore, the total deviation is conservatively bounded by

$$|s_i^\star - s_i'| \le \frac{5}{2}\theta, \qquad |x_{ij}^\star - x_{ij}'| \le \frac{5}{2}\theta.$$

Define the loss difference

$$\Delta\ell_i^{(j)} \;=\; \frac{\left(\hat{x}_{ij}' - x_{ij}'\right)^2 - \left(\hat{x}_{ij}^\star - x_{ij}^\star\right)^2}{\sigma_j^2}.$$

Using $a^2 - b^2 = (a+b)(a-b)$, we rewrite

$$\Delta\ell_i^{(j)} \;=\; \frac{\left[(\hat{x}_{ij}' + \hat{x}_{ij}^\star) - (x_{ij}' + x_{ij}^\star)\right]\left[(\hat{x}_{ij}' - \hat{x}_{ij}^\star) - (x_{ij}' - x_{ij}^\star)\right]}{\sigma_j^2}.$$

From the bounded-residual condition $|\hat{x}_{ij}' - x_{ij}'| < M$ and $|\hat{x}_{ij}^\star - x_{ij}^\star| < M$, we have

$$\left|(\hat{x}_{ij}' + \hat{x}_{ij}^\star) - (x_{ij}' + x_{ij}^\star)\right| \le 2M.$$

For the second factor, by triangle inequality,

$$\left|(\hat{x}_{ij}' - \hat{x}_{ij}^\star) - (x_{ij}' - x_{ij}^\star)\right| \le |\hat{x}_{ij}' - \hat{x}_{ij}^\star| + |x_{ij}' - x_{ij}^\star|.$$

We first bound the attribute difference as $|x_{ij}' - x_{ij}^\star| \le \frac{5}{2}\theta$. Moreover, from $|(\hat{x}_{ij}' - \hat{x}_{ij}^\star) - (x_{ij}' - x_{ij}^\star)| < 2M$, we obtain

$$|\hat{x}_{ij}' - \hat{x}_{ij}^\star| \le |x_{ij}' - x_{ij}^\star| + 2M \le \frac{5}{2}\theta + 2M.$$

Combining the above bounds gives

$$\left|(\hat{x}_{ij}' - \hat{x}_{ij}^\star) - (x_{ij}' - x_{ij}^\star)\right| \le 5\theta + 2M.$$

To account for the additional deviation introduced by midpoint-based construction, we conservatively enlarge the bound to

$$\left|(\hat{x}_{ij} - \hat{x}_{ij}^\star) - (x_{ij}' - x_{ij}^\star)\right| \le 15\theta + 10M.$$

Therefore,

$$\Delta\ell_i^{(j)} \le \frac{(2M)(15\theta + 10M)}{\sigma_j^2} = \frac{30\theta M + 20M^2}{\sigma_j^2}.$$

Thus, for all (i, j),

$$\ell_i^{(j)} - \ell_i^{(j)\star} \le \frac{30\theta M + 20M^2}{\sigma_j^2}.$$

Summing over all i and j yields

$$L - L^\star \le \sum_{j=1}^{m} \frac{n\,(30\theta M + 20M^2)}{\sigma_j^2}.$$

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009